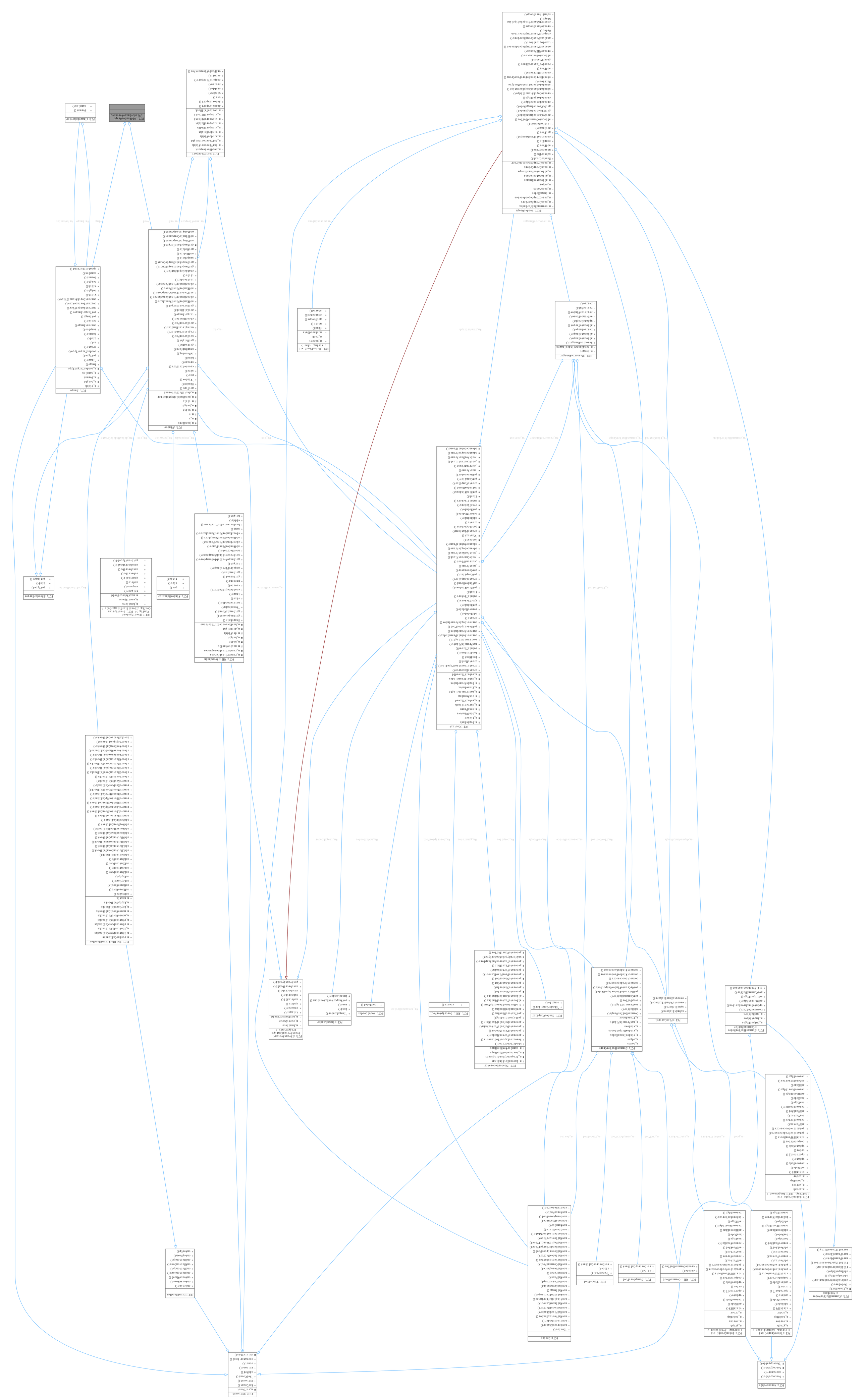

FCT::Noncopyable
+ Noncopyable()
+ operator=()
# Noncopyable()
# ~Noncopyable()
FCT::TokenGraph< std::string, SubmitTicker >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::TokenGraph< std::string, SyncTicker >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::CommandBufferNodes::NodeBase
# m_frameDirty
+ ~NodeBase()
+ updateSynchronization()
+ addOutputEdge()
+ addInputEdge()
+ fillSynchronization()
+ fillAllSynchronization()
+ markFrameDirty()
+ markFrameClean()
+ markAllFramesDirty()
FCT::TokenGraph< std::string, FCT::ImageSaved >
- m_graph
- m_vertex
- m_nodeMap
- m_order
+ visitBFS()
+ addNode()
+ removeNode()
+ update()
+ operator[]()
+ order()
+ updateNode()
+ computeOrder()
+ visitDFSFromRoots()
+ getActivePredecessors()
+ getActiveSuccessors()
- addVertex()
- removeVertex()
- hasVertex()
- addWeakRef()
- removeWeakRef()
- hasEdge()
- hasNode()
- addBoostEdge()
- removeBoostEdge()
- addEdge()
- isZeroRefVertex()
- removeEdge()
FCT::RHI::CommandPool
+ create()
+ createCommandBuffer()
FCT::SemaphorePool
+ alloc()
+ setDestroyCallback()
FCT::FencePool
+ ~FencePool()
+ alloc()
+ setDestroyCallback()
FCT::Device
+ ~Device()
+ newVertexShader()
+ newPixelShader()
+ newRhiVertexShader()
+ newRhiPixelShader()
+ newRhiConstBuffer()
+ newRhiInputLayout()
+ newSingleBufferImage()
+ newMutilBufferImage()
+ newRhiImage()
+ newRhiSwapchain()
+ newRhiPassGroup()
+ newRhiPass()
+ newRhiFence()
+ newRhiSemaphore()
+ newRhiCommandPool()
+ newRhiVertexBuffer()
+ newRhiIndexBuffer()
+ newRhiDescriptorPool()
+ newRhiRenderTargetView()
+ newRhiDepthStencilView()
+ newRhiTextureView()
+ newRasterizationState()
+ newBlendState()
+ newSampler()
+ newPassResource()
+ newSemaphorePool()
+ newFencePool()
+ createResource()
FCT::RefCount
# m_refCount
+ RefCount()
+ RefCount()
+ ~RefCount()
+ addRef()
+ release()
+ count()
+ operator bool()
# deleteThis()
FCT::EventHandler
+ onResize()
+ onMouseMove()
+ onMouseWheel()
+ onLButtonDown()
+ onLButtonUp()
+ onRButtonDown()
+ onRButtonUp()
+ onKeyDown()
+ onKeyUp()
-m_pool
-m_submitTickers
-m_syncTickers
-m_cmdPool
-m_semaphorePool
-m_fencePool
-m_device
FCT::CommandBufferNodes::CommandBuffer
- m_outputEdges
- m_inputEdges
- m_cmdBuffers
+ CommandBuffer()
+ updateSynchronization()
+ addOutputEdge()
+ addInputEdge()
+ getCommandBuffer()
+ fillSynchronization()
FCT::FlowControl
+ submitTickers()
+ syncTickers()
+ executeSubmitTickers()
+ executeSyncTickers()
FCT::CommandBufferGraph
- m_nodes
- m_edges
- m_windowInputNodes
- m_windowOutputNodes
- m_windows
- m_maxFrameInFlight
- m_frameIndex
+ CommandBufferGraph()
+ addBuffer()
+ maxFrameInFlight()
+ swapBuffer()
+ getCommandBuffer()
- getOrCreateWindowInputNode()
- getOrCreateWindowOutputNode()
- connectPredecessors()
- connectSuccessors()
- connectWindowPredecessor()
- connectWindowSuccessor()
FCT::ShaderCompiler
+ ~ShaderCompiler()
+ compile()
FCT::ShaderGenerator
# m_layoutSetBindings
# m_frequencyBindingCount
# m_textureSetBindings
# m_samplerSetBindings
+ ShaderGenerator()
+ ResourceLayoutToElements()
+ generateVertexShader()
+ generatePixelShader()
+ generateDefaultVertexMain()
+ generateDefaultPixelMain()
+ getLayoutBinding()
+ getTextureBinding()
+ getSamplerBinding()
+ findTextureElementByName()
+ allocateTextureBinding()
+ allocateSamplerBinding()
# generateShaderIn()
# generateShaderIn()
# generateShaderOut()
# generateShaderOut()
# generateFromPixelLayout()
# generateVertexMain()
# generatePixelMain()
# generateTexturesAndSamplers()
# uniformTypeToShaderType()
# generateConstBuffer()
FCT::RHI::DescriptorPool
+ create()
#m_resourceDevice
FCT::ModelLoader
+ loadModel()
FCT::ImageLoader
+ ~ImageLoader()
+ load()
+ save()
+ getSupportedExtensions()
# ImageLoader()
FCT::IEventSystem< EventSystemConfig::TriggerOnly >
- m_handlers
- m_eventQueue
- m_nextSubscribeId
+ trigger()
+ enqueue()
+ update()
+ updateAll()
+ subscribe()
+ unsubscribe()
+ unsubscribeAll()
- getEventTypeId()
FCT::CallBackEventHandler
- m_resizeCallbacks
- m_lButtonDownCallbacks
- m_lButtonUpCallbacks
- m_rButtonDownCallbacks
- m_rButtonUpCallbacks
- m_mouseMoveCallbacks
- m_mouseWheelCallbacks
- m_keyDownCallbacks
- m_keyUpCallbacks
- m_nextId
+ onResize()
+ onMouseMove()
+ onMouseWheel()
+ onKeyDown()
+ onKeyUp()
+ onLButtonDown()
+ onLButtonUp()
+ onRButtonDown()
+ onRButtonUp()
+ addResizeCallback()
+ addLButtonDownCallback()
+ addLButtonUpCallback()
+ addRButtonDownCallback()
+ addRButtonUpCallback()
+ addMouseMoveCallback()
+ addMouseWheelCallback()
+ addKeyDownCallback()
+ addKeyUpCallback()
+ removeResizeCallback()
+ removeLButtonDownCallback()
+ removeLButtonUpCallback()
+ removeRButtonDownCallback()
+ removeRButtonUpCallback()
+ removeMouseMoveCallback()
+ removeMouseWheelCallback()
+ removeKeyDownCallback()
+ removeKeyUpCallback()
+ clearResizeCallbacks()
+ clearLButtonDownCallbacks()
+ clearLButtonUpCallbacks()
+ clearRButtonDownCallbacks()
+ clearRButtonUpCallbacks()
+ clearMouseMoveCallbacks()
+ clearMouseWheelCallbacks()
+ clearKeyDownCallbacks()
+ clearKeyUpCallbacks()
+ invokeResizeCallbacks()
-m_dependencyGraph
#m_flowControl
-m_resourceDevice
#m_cmdGraph
#m_compiler
#m_generator
#m_descriptorPool
#m_modelLoader
#m_imageLoader
FCT::Context
# m_logicTask
# m_ticker
# m_bindWindows
# m_nextFrame
# m_currentFlush
# m_submitThread
# m_ctxRunning
# m_maxFrameInFlight
# m_frameIndex
# m_logicFrameIndex
# m_submitFrameIndex
# m_submitThreadId
+ createResource()
+ createTraditionPipeline()
+ createMesh()
+ loadMesh()
+ loadTexture()
+ submitThread()
+ maxFrameInFlight()
+ maxFrameInFlight()
+ currentSubmitFrameIndex()
+ currentFrameIndex()
+ getDescriptorPool()
+ currentLogicFrameIndex()
+ create()
+ addModule()
+ removeModule()
+ getModule()
+ syncTickers()
+ submitTickers()
+ flush()
+ getBindWindows()
+ onWindowBound()
+ createCompiler()
+ getCompiler()
+ getGenerator()
+ _nextFrame()
+ _currentFlush()
+ _waitCurrentFlush()
+ _waitForNextFrame()
+ advanceLogicFrame()
+ advanceSubmitFrame()
# Context()
# ~Context()
# createPlatform()
# postLogicTask()
* create()
* addModule()
* removeModule()
* getModule()
* syncTickers()
* submitTickers()
* flush()
* getBindWindows()
* onWindowBound()
* createCompiler()
* getCompiler()
* getGenerator()
* _nextFrame()
* _currentFlush()
* _waitCurrentFlush()
* _waitForNextFrame()
* advanceLogicFrame()
* advanceSubmitFrame()
-m_flowControl
FCT::RHI::Swapchain
# m_renderFinshFences
# m_renderFinshSemaphores
# m_nativeHandle
# m_width
# m_height
# m_dstWidth
# m_dstHeight
# m_hasRecreatedInThisFrame
+ Swapchain()
+ getImageCount()
+ getSampleCount()
+ ~Swapchain()
+ nativeHandle()
+ size()
+ image()
+ enableDepthBuffer()
+ create()
+ present()
+ getFormat()
+ getSamples()
+ acquireFirstImage()
+ target()
+ getImageAvailableSemaphore()
+ setPresentFinshSemaphore()
+ needRecreate()
+ addRenderFinshFence()
+ clearRenderFinshFences()
+ addRenderFinshSemaphore()
+ clearRenderFinshSemaphores()
+ sync()
+ hasRecreatedInThisFrame()
+ width()
+ height()
-m_resourceDevice
FCT::WindowBehavior
+ pos()
+ size()
+ title()
FCT::IEventSystem< Config >< FCT::EventSystemConfig::IdentifierTriggerOnly >
- m_handlers
- m_eventQueue
- m_nextSubscribeId
+ trigger()
+ enqueue()
+ update()
+ updateAll()
+ subscribe()
+ unsubscribe()
+ unsubscribeAll()
- getEventTypeId()
#m_callbackHandler
FCT::IRenderTarget
+ getType()
+ bind()
+ getImage()
-m_commandBufferToken
-m_flowControl
-m_commandBufferGraph
#m_resourceManager
-m_context
#m_ctx
#m_swapchain
#m_behavior
#m_ctx
#m_delayModuleCreate
FCT::Window
# m_handlers
# m_x
# m_y
# m_width
# m_height
# m_title
# m_needEnableDepthBuffer
# m_depthBufferFormat
+ getType()
+ Window()
+ ~Window()
+ pos()
+ size()
+ createPlatform()
+ create()
+ bind()
+ isRunning()
+ swapBuffers()
+ getWidth()
+ getHeight()
+ setCursorPos()
+ registerHandler()
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+ getCursorPos()
+ clearHandler()
+ targetImage()
+ getCallBack()
+ getCurrentTarget()
+ addRenderFinshSemaphore()
+ clearRenderFinshSemaphores()
+ setPresentFinshSemaphore()
+ addRenderFinshFence()
+ clearRenderFinshFences()
+ initRender()
+ title()
+ enableDepthBuffer()
+ getSwapchainImageCount()
+ getSwapchainSampleCount()
+ swapchain()
+ addModule()
+ getModule()
# getSwapchainTarget()
- addSingleComponent()
- addSingleComponent()
- addSingleComponent()
FCT::Image
# m_width
# m_height
# m_format
# m_samples
# m_renderTargetType
+ Image()
+ ~Image()
+ getType()
+ renderTargetType()
+ create()
+ as()
+ bind()
+ format()
+ samples()
+ currentImage()
+ resize()
+ getImage()
+ getTargetImages()
+ currentTargetView()
+ currentTextureView()
+ currentDepthStencilView()
+ width()
+ height()
+ width()
+ height()
+ format()
+ samples()
+ updateToCurrent()
FCT::ResourceManager
- m_target
- m_needChangeIndexImages
+ ResourceManager()
+ allocateImage()
+ allocateImage()
+ resizeImage()
+ allocateTarget()
- updateGraph()
- addvanceFrame()
- registerWindow()
- resizeSub()
- resize()
#m_renderGraph
FCT::UnionFind< std::string, char >
- m_parent
- m_rank
- m_sharedData
+ find()
+ unite()
+ getGroups()
+ connected()
+ shared()
-m_ctx
-m_resourceManager
-m_passesUnions
#m_autoViewport
-m_wnd
+wnd
+img
#m_image
#m_behavior
FCT::RenderGraph
- m_commandBufferIndex
- m_passGroupBarriers
- m_passGroupDependencies
- m_imageNodes
- m_passNodes
- m_edges
- m_allocatedImages
- m_allocatedPasses
- m_allocatedPassGroups
- m_passGroupOrders
- m_passGroupExecutionOrder
+ RenderGraph()
+ subscribe()
+ unsubscribe()
+ addPass()
+ compile()
+ executeAllPassGroups()
+ getPass()
+ getImage()
- initForSubmit()
- allocateCommandBuffer()
- getOrCreateImageNode()
- getOrCreateImageNode()
- getOrCreateImageNode()
- createTextureEdge()
- createTargetEdge()
- createDepthStencilEdge()
- simulatePassGroupExecution()
- simulateExecutionAndAnalyzeBarriers()
- checkBarriersBeforePassGroup()
- executeBarriers()
- addPass()
- resolveTextureSizes()
- groupPasses()
- allocateResources()
- createRHIPasses()
- analyzePassGroupDependencies()
- topologicalSort()
- analyzePassGroupBarriers()
- computePassGroupExecutionOrder()
- createPassGroups()
- convertShaderStageToPipelineStage()
- submitPassGroup()
FCT::AutoViewport
- m_needReviewport
- m_dstViewportWidth
- m_dstViewPortHeight
- m_windowWidth
- m_windowHeight
- m_viewportWidth
- m_viewportHeight
- m_viewportOffsetX
- m_viewportOffsetY
- m_resizeCallBack
+ AutoViewport()
+ AutoViewport()
+ ctx()
+ window()
+ enable()
+ resize()
+ computeViewport()
+ submit()
+ wndPosToViewportPos()
FCT::OldRenderGraph::WindowImageResource
FCT::ImageBehavior
+ format()
+ samples()